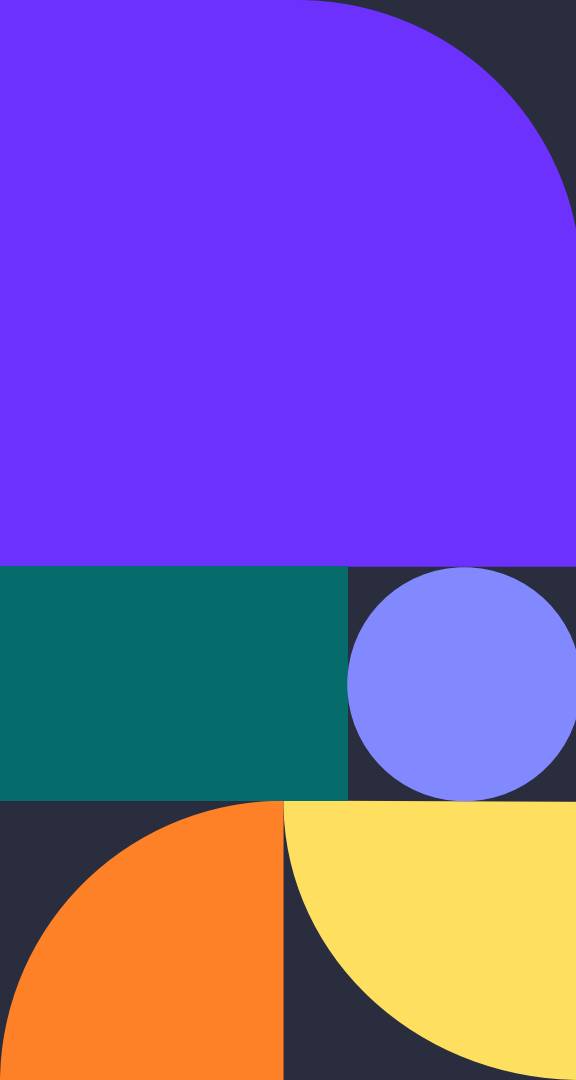


An abstract geometric design on the left side of the slide. It features a large purple quarter-circle in the top-left, a teal rectangle below it, a light blue circle to the right of the teal rectangle, an orange quarter-circle in the bottom-left, and a yellow quarter-circle to the right of the orange one.

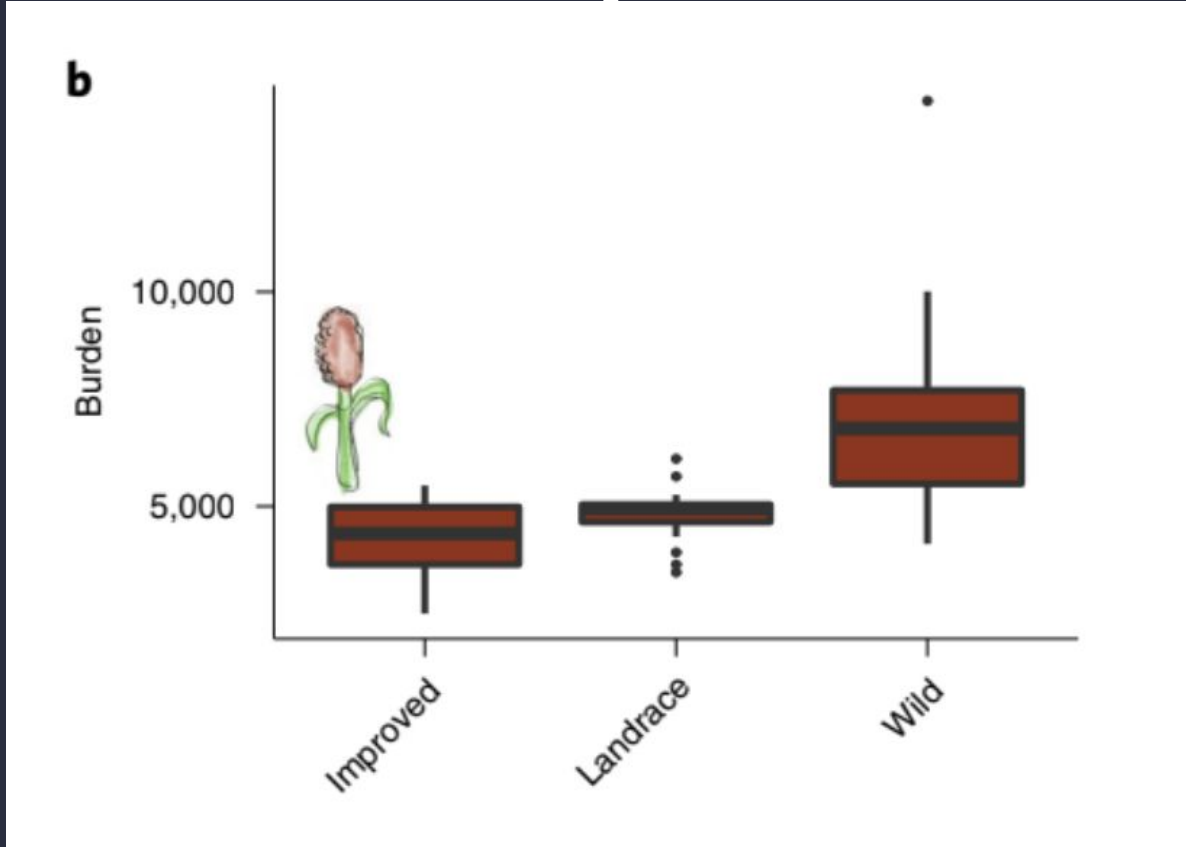
Research Presentation

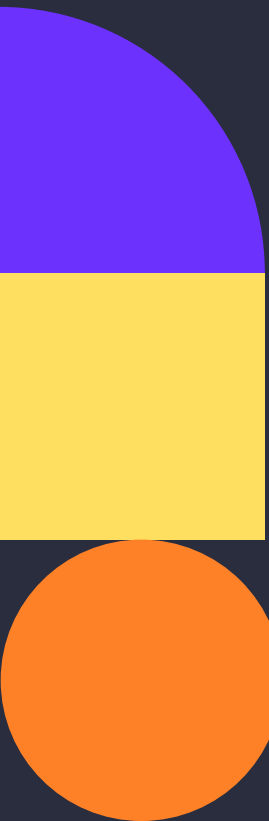
Gloria Jackson

Research Question

Based on what methods you use to determine the ancestral allele, what results do you get?

The Lozano Paper



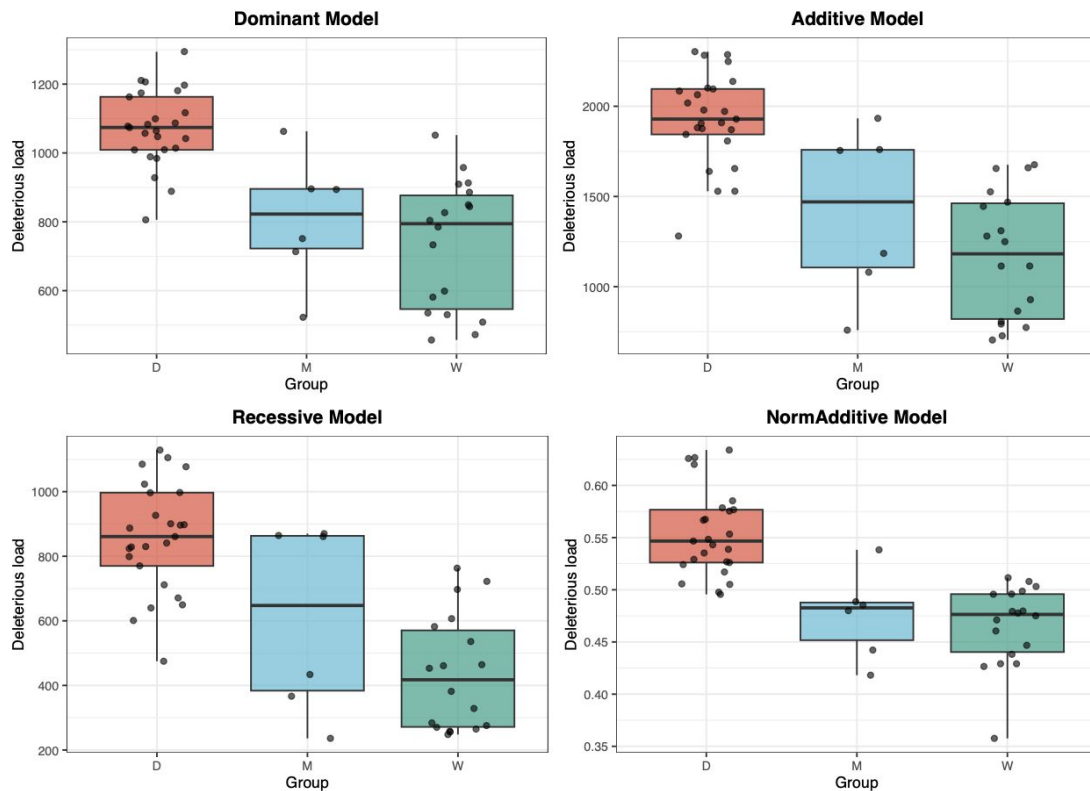


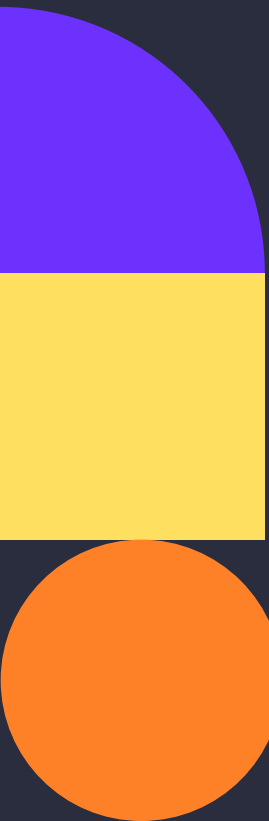
Polarization Methods

01. **Use outgroup as ancestral**
02. **Lozano paper pipeline**

Using the outgroup as the ancestral allele

Deleterious Load Across Models (Excluding Low Confidence Sites)





Polarization Methods

01. **Use outgroup as ancestral**
02. **Lozano paper pipeline**

The Lozano Paper Pipeline

01. **Orthogroups**

02. **Clustal Omega**

03. **TranAlign**

04. **Mapping**

05. **est-sfs software**



Straightforward...or is it?

CSV/TSV files

CDS FASTAs

Nucleotide
alignments

Text files

VCF files

GFF3 files

Protein alignments

Protein FASTAs



Step 0: Organizing Data

Which files do I need?

Focal species: *Sorghum bicolor* (BTx623)

- Protein FASTA
- CDS FASTA
- Gff3 file
- VCF file

Outgroups: *Setaria italica*, Maize

- Protein FASTAs
- CDS FASTAs



Step 0: Organizing Data

Deal Breakers

- Protein FASTAs for each species → NO ASTERISK (TERMINAL CODON)
- ***SEQUENCE HEADERS MUST MATCH IN BOTH PROTEIN AND CDS FASTAS***
 - >Zm00001d033527_P001
 - >Zm00001d033527_T001
 - Trailing '.p' → remove it!!!

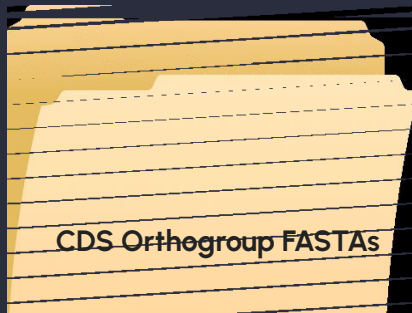
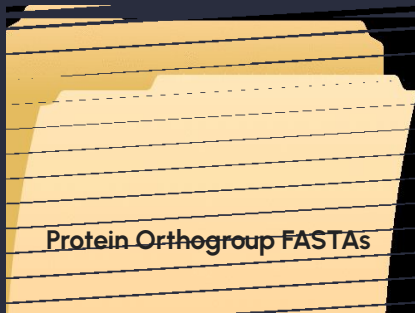


Step 1: Orthogroups

OrthoFinder

Focal species: *Sorghum bicolor* (BTx623)

Outgroups: *Setaria italica*, Maize

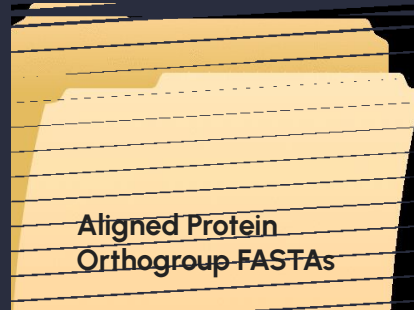




Step 2: Protein Orthogroup Alignments

Clustal Omega

- Go through each protein orthogroup FASTA and align each one with Clustal Omega





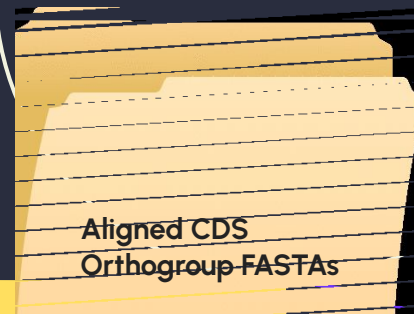
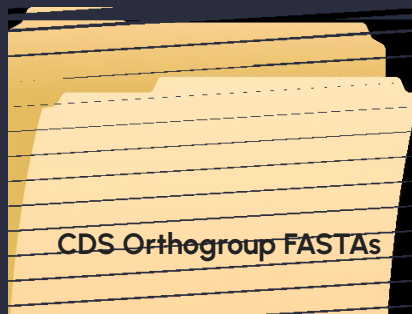
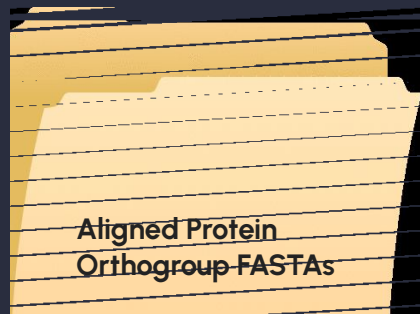
Step 3: Nucleotide Orthogroup Alignment

TranAlign

- Part of the EMBOSS package
- Requires an unaligned nucleotide fasta and an aligned protein fasta

Step 3: Nucleotide Orthogroup Alignment

TranAlign



Step 4: Mapping

Translating alignment positions to their position in the genome

Building the Alignment-to-Genome Mapping

From GFF3 Annotations to Base-Level Genome Coordinates

INPUTS



GFF3 File
Gene models with CDS
exon coordinates



TranAlign Output
CDS-level alignment with
sequence and gaps

Goal: Map each aligned, non-gap base
to its exact genomic position and exon

1 Extract CDS Exon Coordinates

Extract CDS exon coordinates
for the transcript from the
GFF3 file.

chr1 - Exon 1 - Exon 2 - Exon 3 -

GFF3 (CDS features)

SeqID	Source	Type	Start	End	Strand	Attrs
chr1	.	CDS	100	199	199	+
chr1	.	CDS	300	399	399	+
chr1	.	CDS	500	599	599	+
...	...	...	...	...	...	...

2 Sort Exons and Build CDS Coordinate List

Sort exons and build a continuous
list of coding (CDS) genomic
positions.

chr1 - Exon 1 - Exon 2 - Exon 3 -

Sorted Exons

Exon	Start	End	Length
1	100	199	100
2	300	399	100
3	500	599	100

CDS Coordinate List (5' → 3')

100 101 102 ... 199 300 301 ... 399 500 ... 599

Total length = 300 bp

3 Remove Stop Codon

Remove the stop codon
(last 3 bases) from the CDS
coordinate list.

Before

Length = 300 bp

100 101 102 ... 597 599 599

Remove last 3
(stop codon)

After

Length = 297 bp

100 101 102 ... 599

4 Iterate Through the Aligned Sequence

Map each non-gap base to its
genomic position and
corresponding exon.

TranAlign Aligned Sequence
(Example)

Query: A T G - C A G T - T A ...

For each position:
• If base = gap → map to next
CDS genomic position
• If base = gap → no mapping

Mapping in Progress

100 101 102 - 103 104 105 106 - 107 108

(gaps have no genomic position)

5 Output Mapping and Summary Files

Write the base-level mapping
and exon summary files.

Mapping File (Example)

Alignment_Pos	Aligned_Base	Genomic_Pos	Exon
1	A	100	1
2	T	101	1
3	G	102	1
4	-	-	-
5	C	103	1
6	A	104	1
7	G	105	1
...	...	...	...

Exon Summary File (Example)

Exon	Start	End	Strand	Length
1	100	199	+	100
2	300	399	+	100
3	500	599	+	100
...	...	...	...	...

OUTPUTS



Base-Level Alignment → Genome Mapping File
Maps each non-gap aligned base to its genomic position and exon.

Alignment_Pos	Aligned_Base	Genomic_Pos	Exon	Strand
1	A	100	1	+
2	T	101	1	+
3	G	102	1	+
4	-	-	-	-
5	C	103	1	+
6	A	104	1	+
...	...	...	...	...



Exon Summary File
Summary of CDS exons used for the mapping.

Exon	Start	End	Strand	Length (bp)
1	100	199	+	100
2	300	399	+	100
3	500	599	+	100
...	...	...	...	...

What you get

- ✓ A precise map from each aligned base to the genome
- ✓ Exon structure and coordinates used in the mapping

Step 4: Mapping

Plus Strand Example

Exon_Number		Start	End	Strand
1	5360374	5360638	+	
2	5360780	5361237	+	
3	5361362	5361745	+	

Sobic.004G066000_exons.tsv (END)

```
[[gjacks21@str-i3 mapped_coordinates]$ head Sobic.004G066000_mapping.csv  
Base,Alignment_Position,Genomic_Position,Exon
```

```
A,1,5360374.0,1.0
```

```
T,2,5360375.0,1.0
```

```
G,3,5360376.0,1.0
```

```
G,4,5360377.0,1.0
```

```
C,5,5360378.0,1.0
```

```
T,6,5360379.0,1.0
```

```
G,7,5360380.0,1.0
```

```
G,8,5360381.0,1.0
```

```
C,9,5360382.0,1.0
```

```
[[gjacks21@str-i3 mapped_coordinates]$ tail Sobic.004G066000_mapping.csv
```

```
A,1248,5361733.0,3.0
```

```
C,1249,5361734.0,3.0
```

```
A,1250,5361735.0,3.0
```

```
G,1251,5361736.0,3.0
```

```
G,1252,5361737.0,3.0
```

```
G,1253,5361738.0,3.0
```

```
A,1254,5361739.0,3.0
```

```
C,1255,5361740.0,3.0
```

```
A,1256,5361741.0,3.0
```

```
G,1257,5361742.0,3.0
```


Step 4: Mapping

Plus Strand Example: Zooming in on Exons

Exon_Number	Start	End	Strand
1	5360374	5360638	+
2	5360780	5361237	+
3	5361362	5361745	+

Sobic.004G066000_exons.tsv (END)

```
- ,376,,  
- ,377,,  
- ,378,,  
T,379,5360635.0,1.0  
C,380,5360636.0,1.0  
A,381,5360637.0,1.0  
G,382,5360638.0,1.0  
G,383,5360780.0,2.0  
A,384,5360781.0,2.0  
T,385,5360782.0,2.0  
G,386,5360783.0,2.0
```

```
C,874,5361235.0,2.0  
A,875,5361236.0,2.0  
G,876,5361237.0,2.0  
T,877,5361362.0,3.0  
G,878,5361363.0,3.0
```

Step 4: Mapping

Minus Strand Example

Exon_Number	Start	End	Strand
1	61354977	61355051	-
2	61354009	61354212	-
3	61353227	61353484	-
4	61352712	61352927	-
5	61352471	61352626	-
6	61352228	61352387	-
7	61351988	61352139	-
8	61351726	61351813	-
9	61351577	61351650	-
10	61351422	61351496	-
11	61351002	61351334	-

Sobic.002G221700_exons.tsv (END)

```
[gjacks21@str-i3 mapped_coordinates]$ head Sobic.002G221700_mapping.csv
Base,Alignment_Position,Genomic_Position,Exon
A,1,61355051.0,1.0
T,2,61355050.0,1.0
G,3,61355049.0,1.0
G,4,61355048.0,1.0
T,5,61355047.0,1.0
G,6,61355046.0,1.0
G,7,61355045.0,1.0
G,8,61355044.0,1.0
C,9,61355043.0,1.0
[gjacks21@str-i3 mapped_coordinates]$ tail Sobic.002G221700_mapping.csv
C,1920,61351014.0,11.0
T,1921,61351013.0,11.0
C,1922,61351012.0,11.0
T,1923,61351011.0,11.0
A,1924,61351010.0,11.0
T,1925,61351009.0,11.0
T,1926,61351008.0,11.0
G,1927,61351007.0,11.0
G,1928,61351006.0,11.0
A,1929,61351005.0,11.0
```

Step 4: Mapping

Minus Strand Example: Zooming in on Exons

Exon_Number	Start	End	Strand
1	61354977	61355051	-
2	61354009	61354212	-
3	61353227	61353484	-
4	61352712	61352927	-
5	61352471	61352626	-
6	61352228	61352387	-
7	61351988	61352139	-
8	61351726	61351813	-
9	61351577	61351650	-
10	61351422	61351496	-
11	61351002	61351334	-
Sobic.002G221700 exons.tsv (END)			

```
A,122,61354978.0,1.0
G,123,61354977.0,1.0
C,124,61354212.0,2.0
A,125,61354211.0,2.0
A,126,61354210.0,2.0
```

```
A,326,61354010.0,2.0
G,327,61354009.0,2.0
A,328,61353484.0,3.0
A,329,61353483.0,3.0
A,330,61353482.0,3.0
```



Step 5: est-sfs software

Estimated unfolded site frequency spectrum software

- Input files:
 - config-file.txt
 - seed-file.txt
 - data-file.txt





Step 5: est-sfs software

config-file.txt

5.1. Configuration file (**config-file.txt**)

This file (examples provided) consists of three lines of text in the following order:

n_outgroup [1, 2, or 3]
model [0, 1 or 2]
nrandom [0 or positive integer]

model

0 = Jukes-Cantor model
1 = Kimura 2-parameter model
2 = Rate-6 model (see Keightley and Jackson 2018 for details)

nrandom

If nrandom = 0, the ML search algorithm starts with pre-set parameter starting values. If nrandom ≥ 1 , the algorithm returns the highest log likelihood found in nrandom runs using starting values for the parameters randomly sampled from wide ranges. The MLs of the individual runs are reported. In the case of the Rate-6 model, it is recommended that at least 10 random starting value runs are carried out to ensure convergence.



Step 5: est-sfs software

seed-file.txt

5.3. Seed file (**seed-file.txt**)

This text file (example provided) contains a single positive integer. It is overwritten by a new random number seed at the end of the run.





Step 5: est-sfs software data-file.txt

If there are three outgroups, there are 4 space-separated columns. The first column is for the focal species, and the next three columns are for the outgroups. Each column is a comma-separated list of the counts of the four bases in the order A, C, G, T. For example, the first line in the example data file is:

```
20,0,0,0      0,0,0,1 0,0,0,1 0,0,0,1
```

At this site, all $n = 20$ copies sampled in the focal species are A. In the three outgroups, a single copy has been sampled, and in each case it is T. All sites must have the same number of copies sampled in the focal species and up to one copy sampled in each outgroup. If there are missing data in any outgroup, the counts for that outgroup are encoded 0,0,0,0 . Data from polymorphic and non-polymorphic sites are analysed together.

Step 5: est-sfs software

Generating data-file.txt

From Sorghum Variants to Cross-Species Allele Counts

Integrating VCF, Mapping, and Orthogroup Alignments



Input: Sorghum VCF
Multi-sample variants



Gene Mapping Files
Map genomic positions to
alignmant positions



Orthogroup Alignments
Contain sorghum, setaria,
and maize sequences

1. Variant Processing (VCF)

Iterate through each variant in the sorghum VCF

CHROM	POS	ID	REF	ALT	SAMPLES (Genotypes)
1	12345	.	A	G	0/0 0/1 1/1 0/1 ...
1	12398	.	C	T	0/1 0/0 0/1 1/1 ...
...	...	...	...	...	...

Is there a mapping
file for this gene?

No Skip to next site

Yes

Record Variant Metadata
Gene ID, Chromosome, Position,
Ref, Alt

Compute Sorghum Allele Counts
(across ALL samples)



Aggregate counts in A, C, G, T format

Sorghum Counts (example)
(REF = A, ALT = G)

A	C	G	T
12	0	8	0

2. Mapping: Genomic → Alignment

Use the gene's mapping file



Mapping File (example)

Genomic_Position	Alignment_Pos (position N)	Base
12346	106	A
12345	106	A
12350	107	T
...	...	...

Find row where **Genomic_Position**
matches variant position (X)

Record:

- Alignment position (N)
- Base at that genomic position

Example:
Genomic Pos 12345 → Alignment Pos N = 106
Base = A

3. Cross-Species Extraction

Use the orthogroup alignment file



Orthogroup Alignment (example)

	105	106	107	108	...
Sorghum (sicolor)	A	G	T	C	...
Setaria (italica)	A	A	T	C	...
Maize	A	G	C	C	...
...	...	...	...	...	...

Go to alignment position N (from mapping)
Read the base for each species

Setaria Base at N

A

(+1 count)

Maize Base at N

G

(+1 count)

Add to counts (single base observation)

Output: CSV Row per Variant

Gene ID	Chr	Pos	Ref	Alt	Sorghum Counts				Setaria Counts				Maize Counts			
					A	C	G	T	A	C	G	T	A	C	G	T
Sb01g123456	1	12345	A	G	12	0	8	0	1	0	0	0	0	0	1	0

📍 Sorghum: population-level allele counts from VCF | Setaria & Maize: single-base observations from alignments



Step 5: The Program

est-sfs software

- Input files:
 - config-file.txt
 - seed-file.txt
 - **data-file.txt**
- Output files
 - output-file-sfs.txt
 - **output-file-pvalues.txt**





Step 5: est-sfs software

Results

A	B	C	D	E	F	G	H	I	J	K	L	M	N
GeneID	Chromosome	GenomicPos	Ref	Alt	Sorghum Counts	Setaria Counts	Maize Counts	est-sfs probability	Major_Allele	Minor_Allele	est_Ancestral_Allele	est_Ancestral_Confidence	Propq1 Ancestral_Allele
Sobic.001G000900	Chr01	80269	T	C	A:0 C:12 G:0 T:76	A:0 C:0 G:0 T:2	A:0 C:0 G:0 T:2	0.985385	T	C	T	high	T
Sobic.001G000900	Chr01	80344	T	C	A:0 C:8 G:0 T:80	A:0 C:0 G:0 T:0	A:0 C:2 G:0 T:0	0.547193	T	C	T	low	T
Sobic.001G000900	Chr01	80439	C	T	A:0 C:79 G:0 T:9	A:0 C:2 G:0 T:0	A:0 C:2 G:0 T:0	0.981855	C	T	C	high	C
Sobic.001G000900	Chr01	80703	G	A	A:3 C:0 G:81 T:0	A:0 C:0 G:0 T:0	A:0 C:0 G:0 T:0	0.876038	G	A	G	moderate	G
Sobic.001G000900	Chr01	81102	G	A	A:3 C:0 G:87 T:0	A:0 C:2 G:0 T:0	A:0 C:0 G:2 T:0	1	G	A	G	high	G
Sobic.001G000900	Chr01	81103	C	A	A:3 C:87 G:0 T:0	A:0 C:2 G:0 T:0	A:0 C:2 G:0 T:0	1	C	A	C	high	C
Sobic.001G000900	Chr01	81348	G	T	A:0 C:0 G:78 T:12	A:0 C:0 G:0 T:0	A:0 C:0 G:2 T:0	0.969255	G	T	G	high	G
Sobic.001G003500	Chr01	305187	C	G	A:0 C:79 G:13 T:0	A:0 C:2 G:0 T:0	A:0 C:4 G:0 T:0	0.981855	C	G	C	high	C
Sobic.001G008200	Chr01	759670	A	T	A:61 C:0 G:0 T:21	A:0 C:2 G:0 T:0	A:0 C:2 G:0 T:0	0.699708	A	T	A	moderate	T
Sobic.001G008200	Chr01	760099	C	G	A:0 C:63 G:19 T:0	A:0 C:0 G:2 T:0	A:0 C:0 G:0 T:2	0.454372	C	G	G	low	G
Sobic.001G008200	Chr01	761129	A	G	A:73 C:0 G:5 T:0	A:2 C:0 G:0 T:0	A:2 C:0 G:0 T:0	0.930911	A	G	A	high	A
Sobic.001G008200	Chr01	761800	G	A	A:6 C:0 G:78 T:0	A:0 C:0 G:2 T:0	A:0 C:0 G:2 T:0	0.973666	G	A	G	high	G
Sobic.001G008200	Chr01	762219	T	A	A:6 C:0 G:0 T:74	A:0 C:0 G:0 T:2	A:0 C:2 G:0 T:0	0.85484	T	A	T	moderate	T



Step 5: est-sfs software

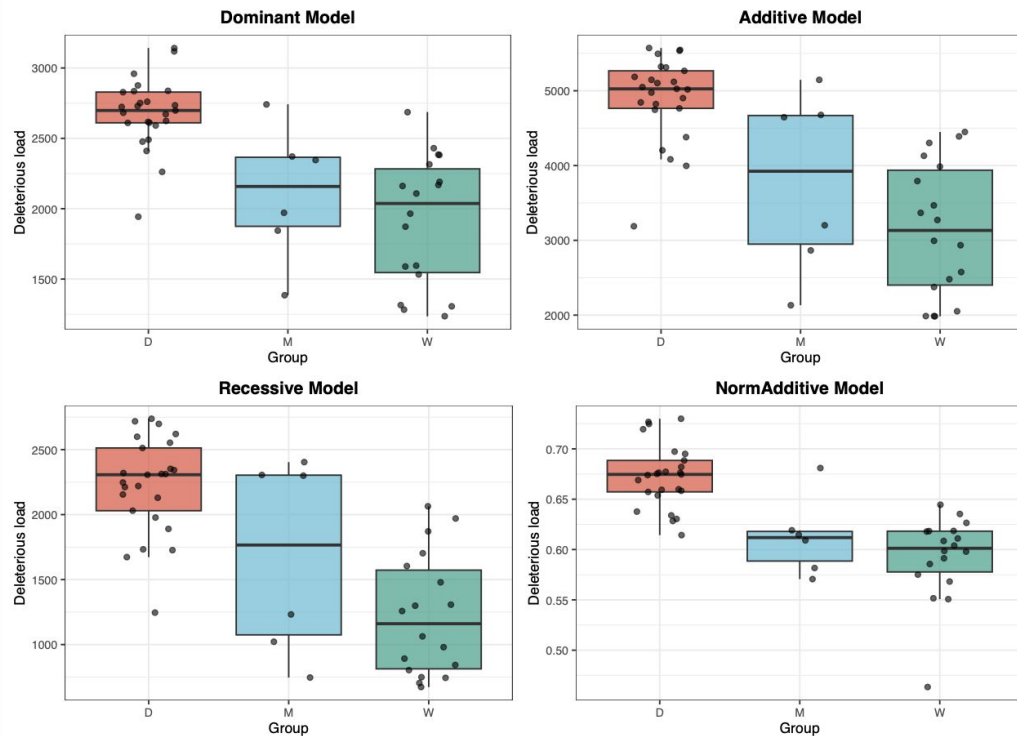
Results

```
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano % python compare_ancestral_alleles.py  
== Ancestral allele comparison ==  
Variants compared : 7673  
Matches           : 5697  
Mismatches        : 1976  
Agreement (%)     : 74.247%
```

Step 5: est-sfs software

Results

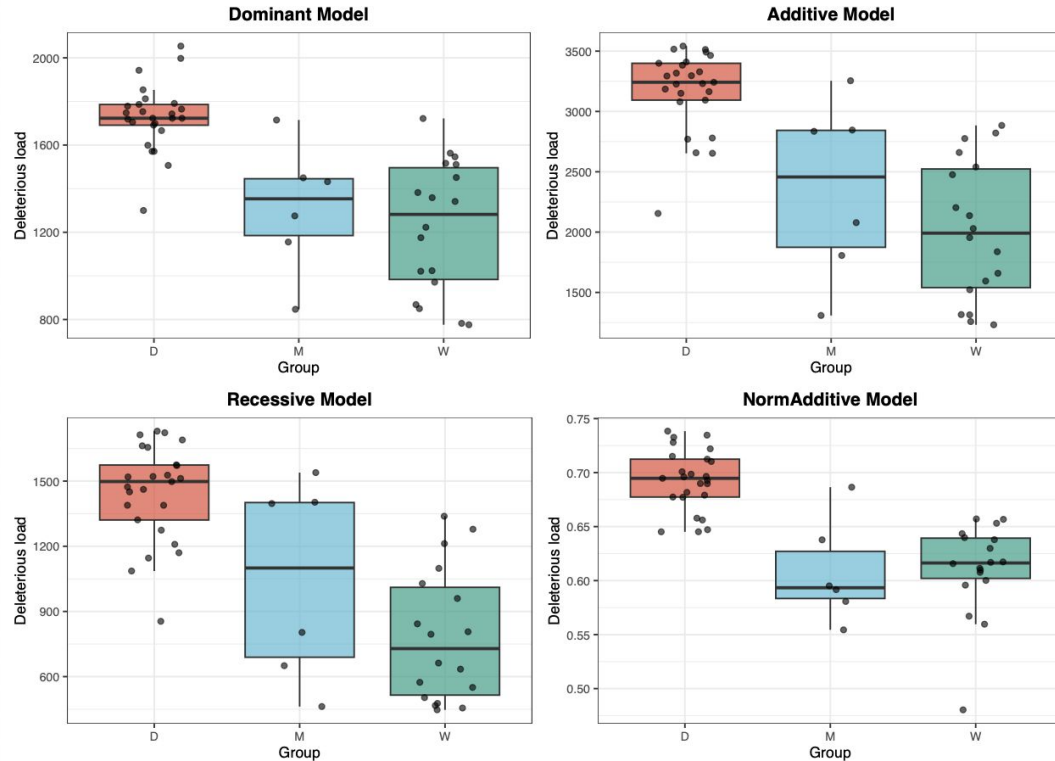
FILTERED RERUN: Deleterious Load Across Models (No Low Confidence Sites)



Step 5: est-sfs software

3 Outgroups

FILTERED 3 OUTGROUPS: Deleterious Load Across Models (No Low Confidence Sites)





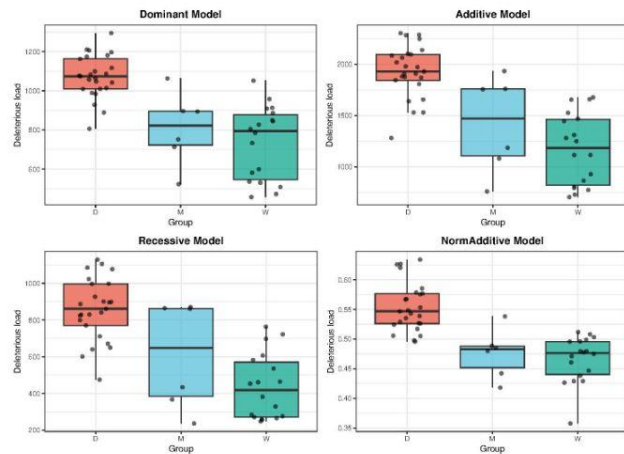
Step 5: est-sfs software

Comparing Results

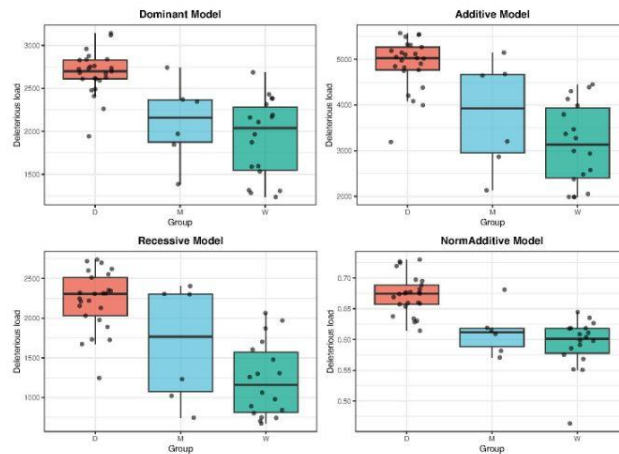
```
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano % less compare_ancestral_calls.py  
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano % python compare_ancestral_calls.py  
Total shared variants: 3453  
Matching variants: 3154  
Disagreement variants: 299  
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano %
```

Put it all together now!

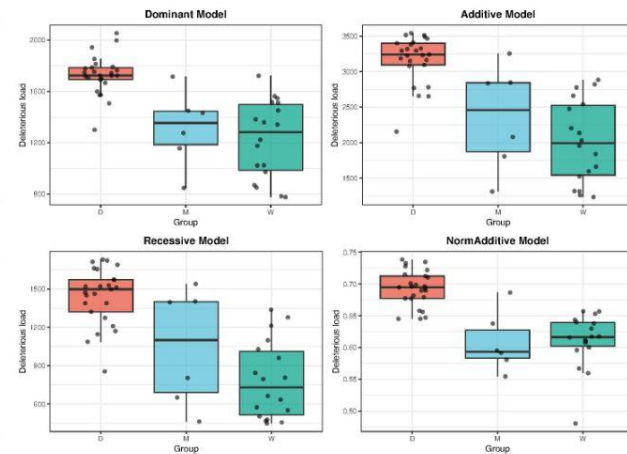
Deleterious Load Across Models (Excluding Low Confidence Sites)



FILTERED RERUN: Deleterious Load Across Models (No Low Confidence Sites)



FILTERED 3 OUTGROUPS: Deleterious Load Across Models (No Low Confidence Sites)



Takeaways

- Went into this thinking polarization might have a big impact on results...it doesn't
- Deleterious load consistently higher in domesticated regardless of polarization method



Thank you :)